

MUSKAN MITTAL

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PROFESSIONAL SUMMARY

Data Professional with 2+ years delivering production-grade data solutions for a US life-sciences firm. Cut data-prep time by **40%**, automated workflows saving **~60% analyst effort per sprint**, and achieved **100% pipeline reliability** under peak load. Expert in SQL, Python, ETL pipelines, and ML-driven data solutions.

PROFESSIONAL EXPERIENCE

Data Scientist | **Mlytics Life Sciences — US (Remote)** | Jul 2024 – Present

- ▶ **Accelerated ML delivery by 40%**: engineered Python + SQL pipelines cutting data-prep time, enabling 3 models to reach production on schedule — eliminating revenue delays.
- ▶ **Reclaimed ~60% analyst capacity per sprint** by automating 5+ reporting & validation workflows with reusable SQL jobs and data-quality checks.
- ▶ **Achieved 100% pipeline uptime under peak load** by building robust error-handling and integrity gates, eliminating all production failures organisation-wide.
- ▶ Translated business requirements into ML-ready data solutions, reducing back-and-forth cycles and ensuring alignment with model performance objectives.

Data Scientist Intern | **Mlytics Life Sciences — US (Remote)** | Feb 2024 – May 2024

- ▶ **Cut model turnaround time by ~35%** by optimising SQL pipelines for ML training jobs — first time the team had quantified pipeline latency.
- ▶ **Eliminated manual reporting** by building real-time inference dashboards, delivering model-performance visibility for the first time.
- ▶ **Improved prediction quality** by resolving 10+ silent backend data inconsistencies and applying feature engineering & preprocessing to affected datasets.

TECHNICAL SKILLS

Languages: Python, SQL, R

Data Analysis & Processing: Pandas, NumPy, Advanced Excel (Pivot Tables, VLOOKUP, Power Query), ETL Pipelines, Data Cleaning & Transformation, Data Wrangling, EDA, Statistical Analysis, Hypothesis Testing, A/B Testing

ML & Advanced Analytics: Scikit-learn, TensorFlow, Keras, PyTorch, Cross-validation, Model Evaluation (Precision/Recall/F1/AUC), Feature Engineering, NLP, spaCy, Generative AI, RAG, Prompt Engineering

Databases & Querying: MySQL, PostgreSQL, SQL Server — advanced JOINS, CTEs, Window Functions, Query Optimisation, Stored Procedures

Visualisation & BI: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Inference Dashboards, KPI Reporting

Cloud & DevOps: Microsoft Azure, AWS (S3, Lambda basics), Flask, FastAPI, REST APIs, Git/GitHub

Soft Skills: Stakeholder Communication, Cross-functional Collaboration, Data Storytelling, Agile/Sprint Delivery

KEY PROJECTS

Skin Cancer Detection System | TensorFlow · CNN · Flask · Azure · Streamlit

- ▶ **Delivered 88% diagnostic accuracy** on a CNN skin lesion classifier (Benign vs. Malignant) using Early Stopping & ReduceLROnPlateau, outperforming baseline models.
- ▶ **Engineered a data augmentation pipeline** (rotation, zoom, flipping) that improved model generalisation by reducing overfitting on limited medical imaging data.
- ▶ **Deployed a production-ready Streamlit dashboard on Azure** for instant lesion predictions, and built Flask REST API endpoints for integration with third-party healthcare platforms.

GitHub: <https://github.com/Muskan29mm/Skin-Cancer-Detection-System-using-CNN>

Live Demo: <https://skin-streamlit-app2-cahtfkd3epa4d9fp.centralindia-01.azurewebsites.net>

CNN Text Classifier | TensorFlow · Keras · spaCy · NLP · Streamlit

- ▶ **Built an end-to-end NLP pipeline** (tokenisation, lemmatisation, NER via spaCy) feeding a CNN text classifier, achieving high-accuracy multi-class text categorisation.
- ▶ **Optimised model architecture** through hyperparameter tuning (filters, kernel sizes, dropout) reducing overfitting while maintaining classification performance.
- ▶ **Deployed as a real-time Streamlit web app** delivering instant classification results with confidence scores for non-technical users.

GitHub: <https://github.com/Muskan29mm/CNN-text-Classifier>

Live Demo: <https://cnn-text-classifier-nhxzqdbvkzrqzpp5amuxx.streamlit.app>

EDUCATION

Master of Computer Applications	Thapar Institute of Engineering & Technology, Patiala, Punjab	2022 – 2024
Bachelor of Science in Information Technology	RIMT University, Mandi Gobindgarh, Punjab	2019 – 2022